Data Analytics and Machine Learning STAT555

"Exploring the Factors Leading to Gestational Diabetes"

Course Instructor: Prof. Dr. Natalia Rojas-Perilla

Semester: Spring 2023/ 2024

Done by:

- Ms. Shamsa Alnasri- 201201405@uaeu.ac.ae
- Mr.Omar Shakir AlFaiadh- 201735167@uaeu.ac.ae
- Ms. Muna Alshamsi -700045181@uaeu.ac.ae

Abstract:

With the increasing rate of Diabetes worldwide, Diabetes became one of the major global health concerns. Much extensive research has been published on diabetes to understand the factors that raise the risk of having diabetes. Additionally, to understand the prevention and management of having it. This project aims to analyze the most significant factors that can contribute to having Gestational Diabetes and how the genetic factor influences it.

مع تزايد معدل الإصابة بمرض السكري في جميع أنحاء العالم، أصبح مرض السكري أحد المخاوف الصحية العالمية الرئيسية. تم نشر العديد من الأبحاث المكثفة حول مرض السكري لفهم العوامل التي تزيد من خطر الإصابة بمرض السكري. بالإضافة إلى ذلك، لفهم الوقاية من الإصابة به وإدارته. يهدف هذا المشروع إلى تحليل أهم العوامل التي يمكن أن تساهم في الإصابة بسكري الحمل وكيفية تأثير العامل الوراثي عليه.

Table of Content

Abstract: 2
Table of Content 3
List of Tables and Figures 4
Introduction: 5
Objective: 5
Research questions: 6
Structure of the study: 6
Chapter one: Dataset Description and Data Analysis: 7
1.1. Section one: Dataset Description. 7
Sample of the dataset: 8
Summary of the data: 8
1.2. Section two: Exploratory data analysis 8
Summary of the data after cleaning the outliers: 9
Chapter Two: Analysis Results and Discussion: 11
2.2.1 Hierarchical Clustering 12
2.2.2 K-Means Clustering 13
2.2.3 Testing 14
2.3. Tree based methods. 14
Chapter Three: Analysis Results and Discussion: 17
3.1. The Findings 17
3.2. The performance of the Model: 18
3.3. Clustering descriptions: 18
3.4. Answers the research questions: 18
3.5. Challenges: 19
Conclusion: 19

List of Tables and Figures

Chapter 1:

Table 1.1: Diabetes Dataset Variables 7

Table 1.2: Original Diabetes Dataset Summary Statistics 8

Figure 1.1: Diabetes Dataset Variables Outliers 9

Table 1.3: Cleaned Diabetes Dataset Summary Statistics 9

Figure 1.2: Diabetes Variables Exploratory Graphs 10

Chapter 2:

Table 2.1: Correlation Analysis Value 11

Figure 2.1: Correlation Analysis 12

Figure 2.2: Hierarchical Clustering 13

Figure 2.3: K-Means Clustering 13

Figure 2.4: Classification or Decision Trees 15

Table 2.2: Decision Tree Result (Importance of Variables) 15

Figure 2.5: Pruned Classification or Decision Tree 16

Table 2.3: Pruned Decision Tree Result (Importance of Variables) 17

Introduction:

Diabetes is defined by increased glucose levels in the blood as the resulting from defects in insulin. In 2023, over 500 million people as the estimation having the Diabetes which is considered one of the most global health widespread. Additionally, predictions indicate that this number will continue to increase.

Diabetes is influenced by many genetic and environmental factors. These factors contribute significantly to disease development. This calls for understanding the relationship between these factors and the extent of their contribution on development of Diabetes.

Diabetes has many diverse types, one of the types is called Gestational Diabetes. This type of Diabetes occurs during pregnancy period when the body can't produce enough insulin to meet the needs of pregnancy which leads to high blood sugar levels.

By analyzing dataset, the study will identify the key determinants of having gestational diabetes and understand the relation between each factor. Additionally, it will contribute to managing gestational diabetes and reducing the probability of having it. Furthermore, reducing the probability of having gestational diabetes by following healthy patterns will help to improve the health outcomes for individuals and community. Through understanding of the factors leading to diabetes, we can work towards preventing and managing these factors effectively.

Objective:

This study aims to explore the relation between some factors in shaping the risk of gestational diabetes and which are the most significant factors that will lead to having it.

Research questions:

- What are the most significant factors associated with the probability of having gestational diabetes?
- How do genetic factors such as family history influence having gestational diabetes?

Structure of the study:

The structure of this study includes the following chapters:

- Chapter one: Dataset Description and Data Analysis:

  The first chapter includes two sections. the first section: provide a comprehensive overview of the dataset used in the analysis. starting by describing the original dataset, including details on data and sources.

  Section two: Exploratory data analysis: provide insights into the dataset structure, variables, and distributions. This section includes also examining the summary statistics, visualization distributions by using different graph such as histograms and box plot. Also, identifying any potential outliers or missing values and any cleaning data. this process to ensure the data quality and integrity.

- Chapter 2: Analysis Methods:

  In the second chapter, will provide in deep analysis phase by employing a variety of statistical and machine learning methods to answer the research questions. This includes decision- tree methods such as classification or regression trees and other parametric methods such as logistic regression, linear regression, and others. These methods allow to model the relationship between predicted variables and the current variables. Additionally, using other analysis such as clustering analysis to identify distinct clusters within the dataset based on similarities and characteristics. By using a diversity of analytical approaches, the comprehensive exploration of the factors contributing to gestational diabetes will be clear.

- Chapter 3: Analysis Results and Discussion

The Final chapter: will provide Overall comprehensive analysis results and addressing the challenges faced. It will include the results of the different analysis, with the findings, comparing the performance of various models, assessing the accuracy, interpretability, and providing detailed descriptions of clusters. After examination of the analysis results, will determine the factors leading to diabetes and the implications in how prevent and manage it.

Chapter one: Dataset Description and Data Analysis:

1.1. Section one: Dataset Description.

The Original source of the dataset is the National Institute of Diabetes and Digestive and Kidney Diseases. The dataset was published in Kaggle, and the sample was taken for pregnant Indian women.

The dataset comprised many factors to find if there are any relations with these factors and having gestational diabetes. The original dataset includes 9 variables with 769 observations as follows:

- Pregnancies: A numerical variable that represents the number of times of being pregnant
- Glucose: A numerical variable that represents plasma glucose concentration 2 hours in an oral glucose tolerance test
- Blood Pressure: A numerical variable that represents diastolic blood pressure (mm Hg)
- Skin Thickness: A numerical variable that represents triceps skin fold thickness (mm)
- Insulin: A numerical variable that shows 2-hour serum insulin (mu U/ml)
- BMI: A numerical variable that represents the body mass index (weight in kg/ (height in m) ^2), which is a measure of body fatness that can be calculated by dividing a person's weight in kilograms by the square of the person's height in meters.
- Diabetes Pedigree Function (refer to the family history): A numerical variable that represents diabetes pedigree function.
- Age: A numerical variable that represents person's age (how many years old)

- Outcome: A categorical variable that can be considered as class variable (0 or 1), where 0 means that a person does not have diabetes while 1 means that a person has diabetes.

Sample of the dataset:

No	Pregnancies	Glucose	Blood Pressure	Skin Thickness	Insulin	BMI	Family History Probability	Age	Outcome
1	1	89	66	23	94	28.1	0.167	21	0
2	0	137	40	35	168	43.1	2.288	33	1
3	3	78	50	32	88	31.0	0.248	26	1
4	2	197	70	45	543	30.5	0.158	53	1
5	1	189	60	23	846	30.1	0.398	59	1
6	5	166	72	19	175	25.8	0.587	51	1

Table 1.1: Diabetes dataset variables

Summary of the data:

Summary of the data **before** cleaning outliers:

	Pregnancies	Glucose	Blood Pressure	Skin Thickness	Insulin	BMI	Family History Probability	Age	Outcome
Min	0.000	0.0	0.00	0.00	0.00	0.00	0.0780	21.00	0.000
1st Quartile	1.000	99.0	62.00	0.00	0.00	27.30	0.2437	24.00	0.000
Median	3.000	117.0	72.00	23.00	30.5	32.00	0.3725	29.00	0.000
Mean	3.845	120.9	69.11	20.54	79.8	31.99	0.4719	33.24	0.349
3rd Quartile	6.000	140.2	80.00	32.00	127.2	36.60	0.6262	41.00	1.000
Max	17.000	199.0	122.00	99.00	846.0	67.10	2.4200	81.00	1.000

Table 1.2: original Diabetes dataset summary statistics

1.2. Section two: Exploratory data analysis

The dataset has gone through many processes to ensure the result does not have any bias. The data set contained 377 missing values, which accounted for 49% of the original dataset. Ensuring the accuracy of the model, 377 observations have been removed from the dataset. To improve the quality of the dataset by examining the outliers using R software to detect outliers, the following box pilots identified the outlier's for each variables .

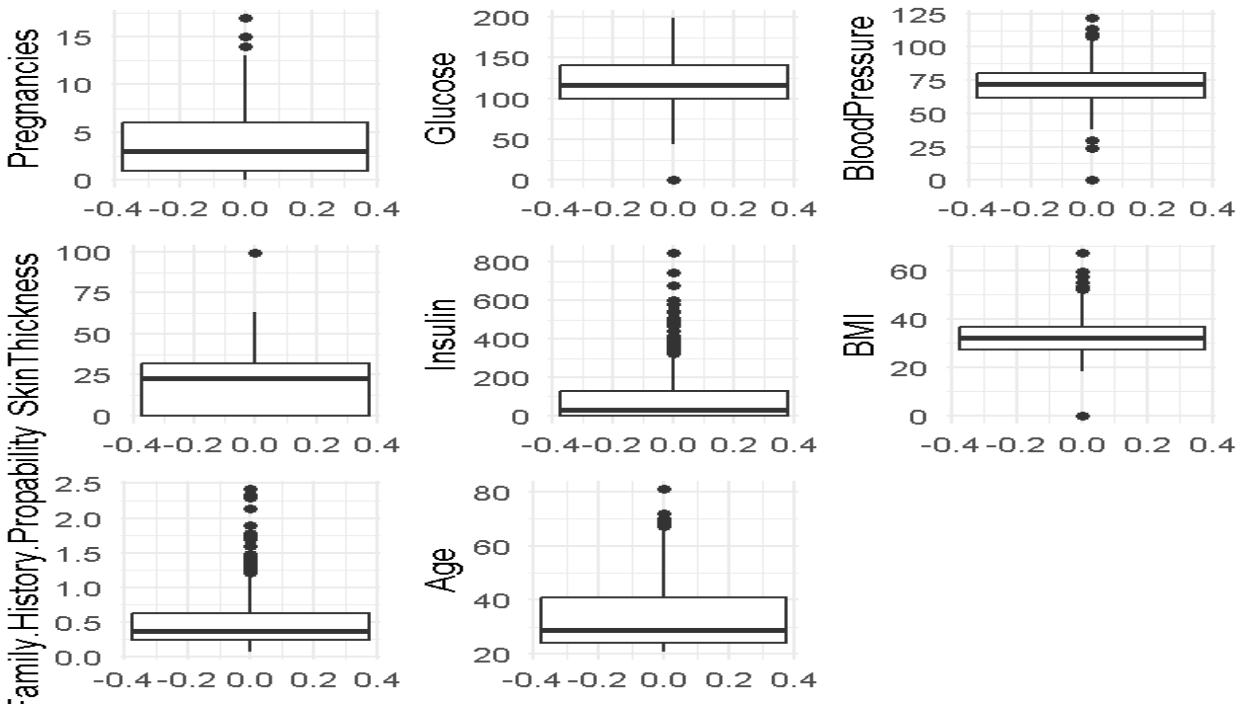


Figure 1.1: Diabetes variables' outliers

The variable "Diabetes Pedigree Function" was renamed to be "Family History Probability" The reason is due to lack of clarity which may lead to misunderstanding and the alternative name "Family History Probability" was chosen after we explored the data, and we understand the variable talks about family history. The cleaned dataset had 392 observations and 9 variables.

Summary of the data after cleaning the outliers:

	Pregnancies	Glucose	Blood Pressure	Skin Thickness	Insulin	BMI	Family History Probability	Age	Outcome
Min	0.000	56.0	24.00	7.00	14.00	18.20	0.0850	21.00	0.0000
1st Quartile	1.000	99.0	62.00	21.00	76.75	28.40	0.2697	23.00	0.0000
Median	3.000	119.0	70.00	29.00	125.50	33.20	0.4495	27.00	0.0000
Mean	3.845	122.0	70.66	29.15	156.06	33.09	0.5230	30.86	0.3316
3rd Quartile	6.000	143.0	78.00	37.00	190.00	37.10	0.6870	36.00	1.0000
Max	17.000	198.0	110.00	63.00	846.00	67.10	2.4200	81.00	1.0000

Table 1.3: Cleaned Diabetes dataset summary statistics.

After cleaning data, we can look at some scatter plots, a bar chart, and histograms. The histogram of Glucose shows that there is no gap after cleaning the missing values or the 0s from the data. The histogram of BMI shows better distribution than the one before cleaning the data where the gap that 0s were causing was removed. The first scatter plot shows the relationship between the number of pregnancies and the age of a person. We can see a positive relationship between these two variables where when a person's age increases, the number of pregnancies increases too. This information will be useful when we reach the part of correlation. The second scatter plot shows the relationship between the family history probability and BMI. We can see that there is a weak relationship between these two variables. This information would be useful in answering our second question about family history probability which we will go through later.

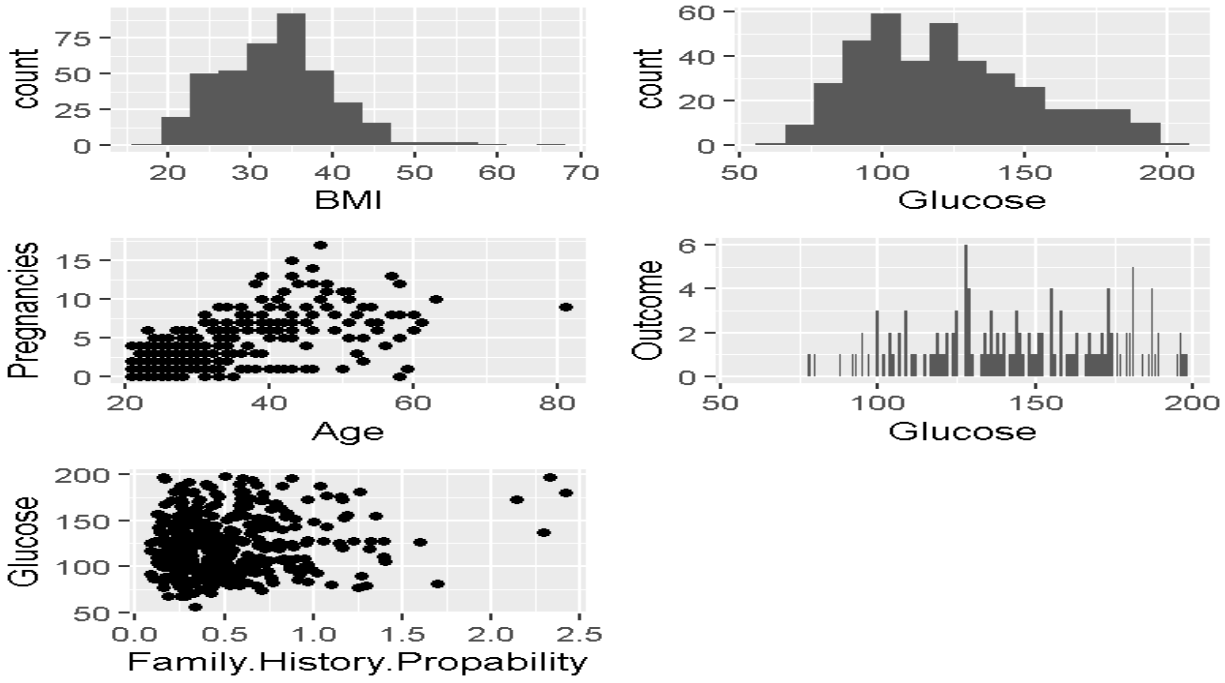


Figure 1.2: Diabetes variables' Graphs

Chapter Two: Analysis Results and Discussion:

This project aims to find out the factors that affect pregnant women and may cause them to end up having gestational diabetes. To help in this task, the dataset was analyzed in both supervised and unsupervised method. In supervised methods, we have used linear regression and classification trees and in unsupervised methods, we have used hierarchical and k-means clustering. Through the analysis of multiple different types of machine learning models, we will be able to compare between the models and choose the one most suitable to help us achieve our target.

2.1 correlation analysis

Before the creation of the machine learning models, a correlation analysis was used on all the variables against the target variable, Outcome. The highest correlation of variables against the Outcome were with Glucose and Age at 0.5156027 and 0.3508038 respectively. While the lowest correlation was with Blood Pressure and Family History at 0.1926733 and 0.2093295 respectively.

Variable X Outcome	Correlation Score
Pregnancies X Outcome	**0.256566**
Glucose X Outcome	**0.5156027**
Skin Thickness X Outcome	**0.2559357**
Insulin X Outcome	**0.3014292**
Blood Pressure X Outcome	**0.1926733**
Family History X Outcome	**0.2093295**
BMI X Outcome	**0.2701184**
Age X Outcome	**0.3508038**

Table 2.1: correlation analysis value

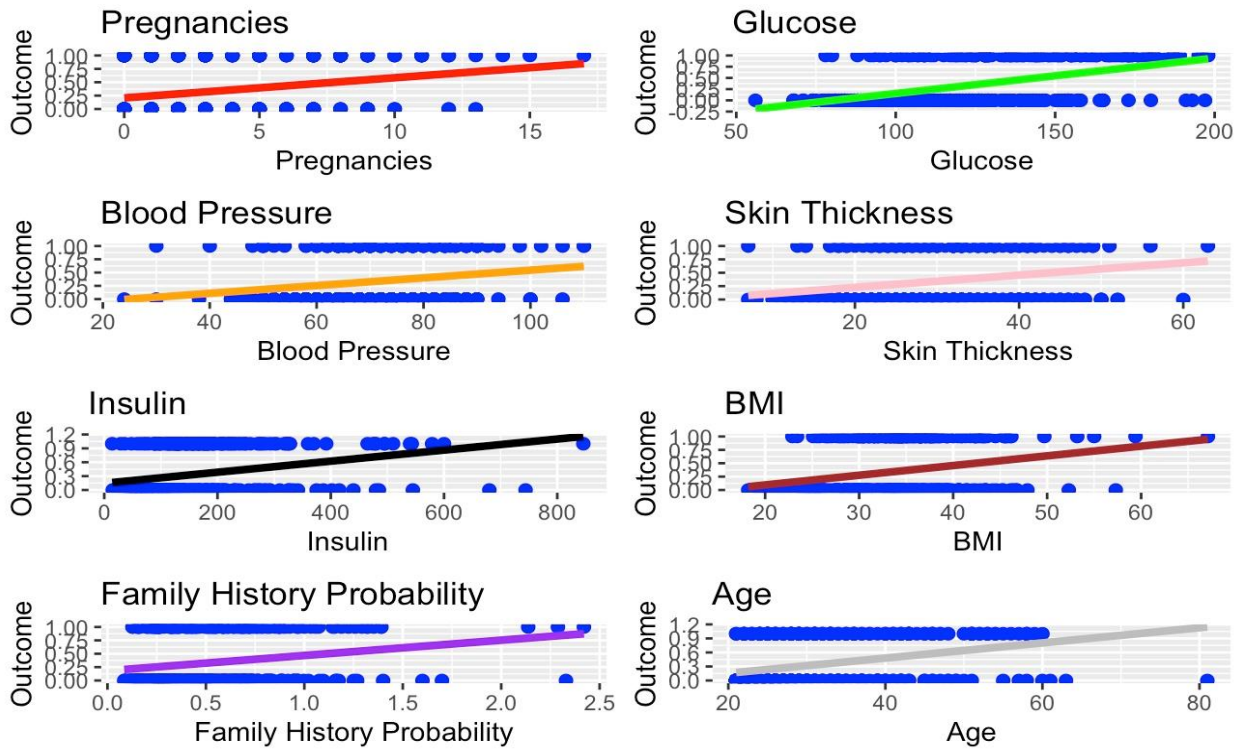


Figure 2.1: correlation analysis

2.2. Clustering

Clustering methods are unsupervised machine learning models that will group data based on their common attributes, and in our analysis, we have used two clustering related machine learning models: Hierarchical Clustering and K-Means Clustering.

2.2.1 Hierarchical Clustering

Hierarchical clustering is a diagram that is very similar to a decision tree, however, its main purpose is to group similar data points together. In the analysis, our data was divided into 4 (K = 4) clusters by R software. However, due to the large range between the data points, the dendrogram wasn't very clear and it was not easy to understand, as shown in *Figure 2.2: Hierarchical Clustering*. Clustering was made; however, the selection of the clustering groups was not clear. Thus, we conducted K- means clustering as well.

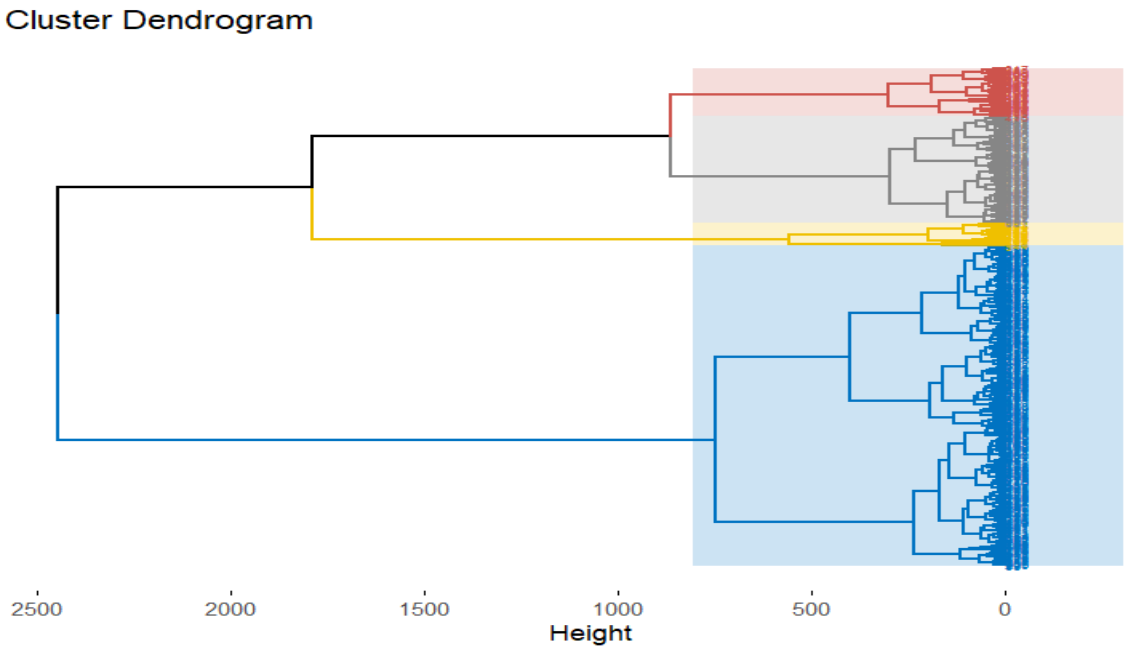


Figure 2.2: Hierarchical Clustering

2.2.2 K-Means Clustering

K-Means clustering is a machine learning model that is usually used for a large amount of data. In the analysis, our data was clustered into 4 clusters, However, due to the large range between the data points, the result wasn't very clear, and it indicated an overfitted model.

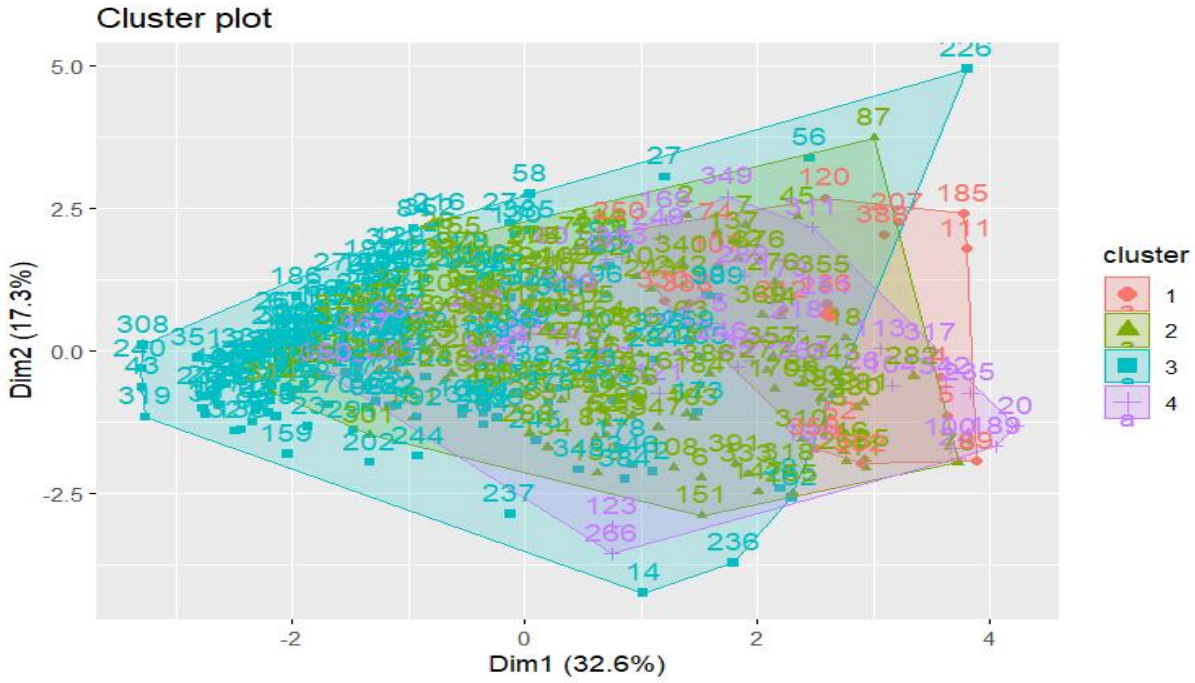


Figure 2.3: K-Means Clustering

2.2.3 Testing

Cross validation was used to test the model, and specifically using K-Fold. The data is segmented in equal k parts, these k parts are then put into the model one by one, while the rest of the parts are used as testing data and the test is repeated k number of times. However, the result of the testing was larger than expected, 6371072. A larger number in the testing means that the clustering is not able to make very clean clusters or separate the data points in clusters as effectively as intended, but instead, there is a case of overlapping which is visually evident and is most likely due to the very large range of data points in the dataset.

2.3. Tree based methods.

Classification or decision trees are supervised machine learning models that are popular because they're easy to explain, easy to understand and to some people, they are considered to mimic the human thought process. Due to the segmentation of the data when it's put in this machine learning model, we will be able to get a clearer idea of what are the more important variables in this question and hopefully be able to predict the possibility of getting gestational diabetes for any new patient.

As the classification tree relies on choosing between binary categories, our target variable was chosen accordingly:

"Outcome" with the categories meaning having gestational diabetes or not:

- 0: No gestational diabetes
- 1: Has gestational diabetes

The resulting decision tree shows the effect of the variables on the possibility of having gestational diabetes:

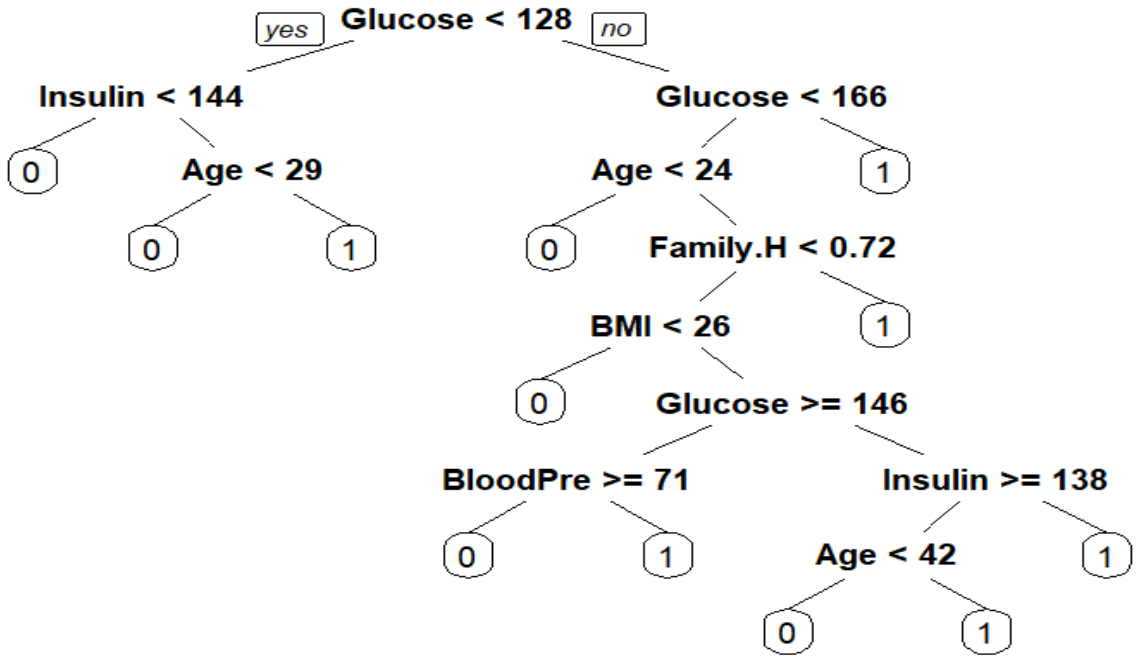


Figure 2.4: Classification or decision trees

To better understand the importance of each variable, we tested the importance of each of them according to the produced machine learning model and got very similar results to the correlation analysis. The most important variables were Age and Glucose at 64.67585 and 54.50380 respectively. While the least important variables were Blood Pressure at 10.19292 and Family History at 19.11424.

Variable	Overall Importance
Age	**64.67585**
Blood Pressure	**10.19292**
BMI	**41.51433**
Family History	**19.11424**
Glucose	**54.50380**
Insulin	**34.43477**
Pregnancies	**31.55310**
Skin Thickness	**19.35930**

Table 2.2: Decision tree result (Importance of Variables)

To make sure that we have the optimal machine learning model, we opted to prune the tree and in turn, got a much smaller tree that emphasizes the importance of Glucose and Age when it comes to the possibility of pregnant women having gestational diabetes.

After pruning the tree to make the features based on "Glucose" and "Age" show that people with Glucose less than 128 tend to "Not have diabetes". People with Glucose with more than 166 tend to "have" diabetes". People with Glucose between 128 and 166 and Age less than 24 tend to "Not have diabetes". While people with Glucose between 128 and 166 and Age more or equal to 24 tend to "have diabetes".

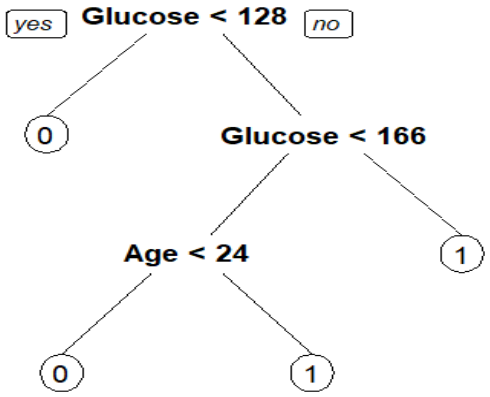


Figure 2.5: pruned Classification or decision trees

Considering that it's a much smaller machine learning model, we tested the importance of variables once again to better understand the relationship between them and the outcome in the pruned tree. The results were similar to the correlation analysis and the importance of variables in the full decision tree model. The most important variables were Glucose at 51.122415 and Age at 41.893661. While the least important variables were Family History and Blood Pressure at 2.925346 and 0.000000 respectively.

Variable	Overall Importance
Age	**41.893661**
Blood Pressure	**0.000000**
BMI	**28.273272**
Family History	**2.925346**
Glucose	**51.122415**
Insulin	**25.038043**
Pregnancies	**21.183752**
Skin Thickness	**11.717913**

Table 2.3: pruned Decision Tree result (Importance of Variables)

Chapter Three: Analysis Results and Discussion:

3.1. The Findings

Based on the results detailed on the chapter two and after applying verity analysis methods to answer the research questions, the comprehensive analysis results are:

- correlation analysis results indicate the correlation between all the variables against the target variable (Outcome). The results show the highest correlation of variables against the Outcome were with Glucose and Age at 0.5156027 and 0.3508038 respectively. While the lowest correlation was with Blood Pressure and Family History at 0.1926733 and 0.2093295 respectively.
  The analysis answers the first research question "What are the most significant factors associated with the probability of having gestational diabetes?" Glucose and Age are considered the most significant factors based on the correlation analysis results. While genetic factors such as Family History indicate that there is a correlation, it was not significant.
- Classification or decision tree results show the most important variables which were Age and Glucose with 64.67585 and 54.50380 respectively. While the least important variables were Blood Pressure at 10.19292 and Family History at 19.11424. After pruning the tree to make the features based on "Glucose" and "Age" show that ***people with Glucose less than 128 tend to "Not have diabetes". People with Glucose with more than 166 tend to "have" diabetes". People with Glucose between 128 and 166 and Age less than 24 tend to "Not have diabetes". While people with Glucose between 128 and 166 and Age more or equal to 24 tend to "have diabetes".***

The analysis answers the first research question "What are the most significant factors associated with the probability of having gestational diabetes? " . Glucose and Age considering the most significant factors based on the tree analysis results. While the genetic factors such as family history indicates that there is a low significant.

3.2. The performance of the Model:

The Cross validation was used to test the model, using K-Fold. The data is segmented in equal k parts. The result of the testing was larger than expected, 6371072. A larger number in the testing means that the clustering is not able to make very clean clusters or separate the data points in clusters as effectively as intended, but instead, there is a case of overlapping which is visually evident and is most likely due to the very large range of data points in the dataset.

3.3. Clustering descriptions:

- Hierarchical Clustering results divided the data into 4 clusters. However, due to the large range between the data points, the dendrogram wasn't very clear nor was it easy to understand.
- K-Means clustering results clustered the dataset into 4 clusters, However, due to the large range between the data points, the result wasn't very clear, and it indicated an overfitted model.

3.4. Answers the research questions:

- What are the most significant factors associated with the probability of having gestational diabetes? **Based on the correlations results and the decision tree result, Glucose and Age are considered the most significant factors that lead to having gestational diabetes.**
- How do genetic factors such as family history influence having gestational diabetes? **Based on the correlations results and the decision tree result, genetic factors such as family history has low influence that led to have gestational diabetes.**

3.5. Challenges:

- The dataset contains a lot of missing values and outliers, which indicate a lack of accuracy and makes it not suitable for the usual clustering or prediction methods.
- Based on analysis, the resulting clusters using the hierarchical and K-means methods are suffering from overfitting and have poor results when applied to the data.

Conclusion:

Referring to the analysis results and based on the challenges, the dataset which was used does not support the project completely and it needs deeper investigations on the factors (variables), considering different characteristics with more observations. Further analysis could be used in exploring the characteristics of each variable and clusters in details. By using clustering results and other analyzing methods, the results will support appropriate preventive treatments, and continuous monitoring to assist in predicting the risk of gestational diabetes.